

Introduction

When a bar chart needs to display two categorical variables at once, three layout options compete: stacking the sub-bars within each main bar (showing totals and composition), placing them side by side (showing within-category comparison), or normalising to 100 % (showing proportions independent of total). Each layout answers a different question, and choosing the right one is more important than tweaking colours or fonts. The `position` argument of `geom_bar()` and `geom_col()` selects between the three.

Prerequisites

A working knowledge of bar charts, the distinction between `geom_bar()` (counts) and `geom_col()` (pre-computed values), and `ggplot2` aesthetic mappings.

Theory

`position = "stack"` is the default for bar geoms with a fill aesthetic: sub-bars stack vertically inside each main bar, the bar's total height equals the sum of the sub-counts, and segments encode composition.

`position = "dodge"` (or `position_dodge(width = ...)` for finer control) places sub-bars side by side, one per fill level, at the same x-tick. Each sub-bar's height equals its own count, making within-x comparison direct.

`position = "fill"` stacks but normalises every main bar to height 1 (or 100 %), hiding the absolute totals to expose proportional structure across the x-axis.

The choice maps directly onto the analytical question: “How big is each group?” → stack; “How does subgroup A compare to subgroup B within each main category?” → dodge; “How does the proportion of subgroup A shift across main categories?” → fill.

Assumptions

Two categorical variables (or one categorical and one ordered) and a count or summary that makes sense to add across sub-bars.

R Implementation

```
library(ggplot2)

d <- data.frame(
  stage = factor(rep(c("I", "II", "III"), 2)),
  sex    = factor(rep(c("F", "M"), each = 3)),
  count  = c(30, 45, 25, 25, 50, 30)
)

# Stacked
ggplot(d, aes(stage, count, fill = sex)) +
  geom_col(position = "stack") +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal()

# Dodged
ggplot(d, aes(stage, count, fill = sex)) +
  geom_col(position = position_dodge(width = 0.8)) +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal()
```

```
# Filled (proportions)
ggplot(d, aes(stage, count, fill = sex)) +
  geom_col(position = "fill") +
  scale_y_continuous(labels = scales::percent_format()) +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal()
```

Output & Results

Three views of the same disease-stage-by-sex data: a stacked bar showing the total cases per stage decomposed by sex, a dodged bar enabling direct sex-vs-sex comparison within stage, and a filled bar showing the female-to-male proportion balance across stages with the y-axis as a percentage.

Interpretation

A reporting sentence (figure caption): “Stacked bars show total case counts per stage decomposed by sex; dodged bars display male and female counts side by side for direct comparison within each stage; filled bars normalise to 100 % to expose the sex proportion across stages.” Always describe which layout was chosen and why; it is the most common source of bar-chart misreading.

Practical Tips

- Stacked bars communicate totals cleanly but obscure within-stage comparisons because the segments above the first sub-bar do not start at zero — readers cannot judge their lengths accurately.
- Dodged bars need adequate bar width and stage spacing; with three or more fill levels, the figure widens quickly and benefits from `coord_flip()` for horizontal layouts.
- Filled bars hide absolute counts entirely; pair them with a text or table summary giving the per-stage n .
- For more than four fill levels, stacking and dodging both become hard to read; consider a heatmap, a faceted small-multiple, or a Sankey diagram instead.
- Order fill levels deliberately (`forcats::fct_relevel()`); the legend reads top-to-bottom and the bar reads bottom-to-top, so the visual flow depends on factor order.
- Use `geom_text(aes(label = count), position = position_stack(vjust = 0.5))` to label segments with their counts when the total alone is not enough.

R Packages Used

`ggplot2` for `geom_bar()`, `geom_col()`, and the `position_*` family; `forcats` for factor reordering of the fill levels; `scales` for percent and comma label formatters; `ggalluvial` for flow-style alternatives when stacked bars at three or more levels become confusing.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- `ggplot2` grammar and multi-panel layouts